

3. Mid-term Project (Week 3):

Project Title:

Building a Basic AI Knowledge Assistant with Memory

Project Description:

In this mid-term project, you will develop a functional AI assistant capable of retrieving and answering questions from structured and unstructured knowledge sources. Using modern AI frameworks like LangChain, the assistant will interact with multiple data sources such as PDFs, Google Sheets.

In addition to retrieval capabilities, the assistant will maintain conversational memory, allowing it to remember previous interactions and provide context-aware responses across a full chat session.

The project involves the following key steps:

1. Data Integration:

- Upload and process content from: PDFs , Google Sheets
- Convert data into structured text suitable for downstream processing.

2. Embedding & Indexing:

- Generate vector embeddings using models like HuggingFace or OpenAI embeddings
- Store embeddings in a vector database (e.g., FAISS)
- Enable efficient semantic search over the knowledge base.

3. Question Answering Pipeline

Build a Retrieval-Augmented Generation (RAG) pipeline:

- Accept user queries
- Retrieve relevant documents
- Generate accurate answers using an LLM

4. Memory Integration

- Implement chat memory to store full conversation history.
- Use memory techniques such as:
 - Buffer memory (full chat history)
 - Optional:Buffer Summary memory for long conversations
- Ensure the assistant:
 - Remembers previous user questions
 - Uses past context in future responses
 - Handles follow-up questions correctly

5. Testing & Evaluation:

Validate the assistant's performance by asking questions and measuring response accuracy and relevance.